



ALMA MATER STUDIORUM  
UNIVERSITÀ DI BOLOGNA

# Cybersecurity

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**University of Bologna**, Department of Computer Science and Engineering

- ***Cybersecurity*** (1<sup>st</sup> semester)
- ***Scalable and Reliable Services*** (2<sup>nd</sup> semester)

**Bologna Business School**, ***Cybersecurity Management***

**Johns Hopkins University**, School of Advanced International Studies (SAIS, Bologna), ***Conflict Risk and Stability in Cyberspace***

Director of several masters and courses, e.g.,

- “Cyber Academy” for training of ethical hackers (2015 - 2020)
- Master on “Information security and Law” (2002 - 2012)
- Master on “Digital forensics” and Master on “Cyber Defense Governance” for the Ministry of Defense (2012 - 2020)
- “CyberLab” with the Tel Aviv University (2016 - 2019)



# The course

**Classroom lessons:** 64 hours

**Language:** English

## **Main objective**

*In a digital world, every activity is vulnerable to cyber attacks. At the end of the course the students can know and evaluate the most dangerous cyber threats to public and private organizations. Moreover, they are expected to be able to design, build and manage cybersecure systems and applications by adopting modern preventive and defensive methodologies, strategies and technologies including those based on AI*

## **Prerequisites**

- Fundamental courses characterizing any Bachelor in Computer Science or Computer Engineering: systems, networking, software
- Knowledge about traditional approaches to information security (authentication mechanisms, secure protocols, firewalls) may help



# Calendar - Teaching

**Tuesday - 15:15-17:45, Room E1**

**Wednesday - 15:15-17:45, Room E1**

**REAL HOURS**

*Exact schedule might be subject to variations  
because of expected and unexpected events.*

***Check advices on Virtuale***

**Timetable overlap is a well-known and unsolvable problem.  
Cybersecurity is a course chosen by:**

- **Computer Science**
- **Computer Engineering**
- **Artificial Intelligence**
- **Erasmus students**

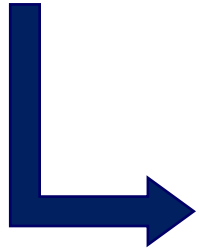
**OUR ONLY  
CHANCE**



# Calendar - *Teaching*

## Scheduled interruptions:

- October 20<sup>th</sup> (Exams in Banca d'Italia)
- October 21<sup>st</sup> (Exams in Banca d'Italia)
- October 28<sup>th</sup> (Laurea degrees)



*Teaching is expected to finish by **December 9<sup>th</sup>***

*You have time to prepare and pass the exam in **December** in the week before Christmas*

**All teaching material will be posted on Virtuale**



# Agenda – *What this course is about*

## Defensive course

1. Digital scenarios and cyber strategies
2. A top-down approach to cybersecurity
3. Cyber threats
4. Information security
5. Cyber risk analysis
6. Qualitative and quantitative cyber risk evaluation
7. Management of cyber risks
8. Technologies and methodologies for cybersecurity (“the fundamental principles”)
9. Assessment and validation methods
10. Innovative cybersecurity solutions
  - Next Generation Firewalls
  - Zero Trust Architectures
  - Cloud security
  - AI for cybersecurity

# A note about the acronyms of this course

**93470** – Computer Science / AI

**97430** – Computer Engineering – Type D  
(*A libera scelta*)

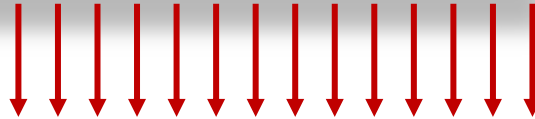
**B5793** – Computer Engineering – Type B  
(*Caratterizzante*)



# The entire “cybersecurity game” is here

**THREATS**

**Threat capability** (skill and resources) of the *cyber threat actor* (or *attacker*, never hacker)



**Vulnerabilities**

**Resistance strength**

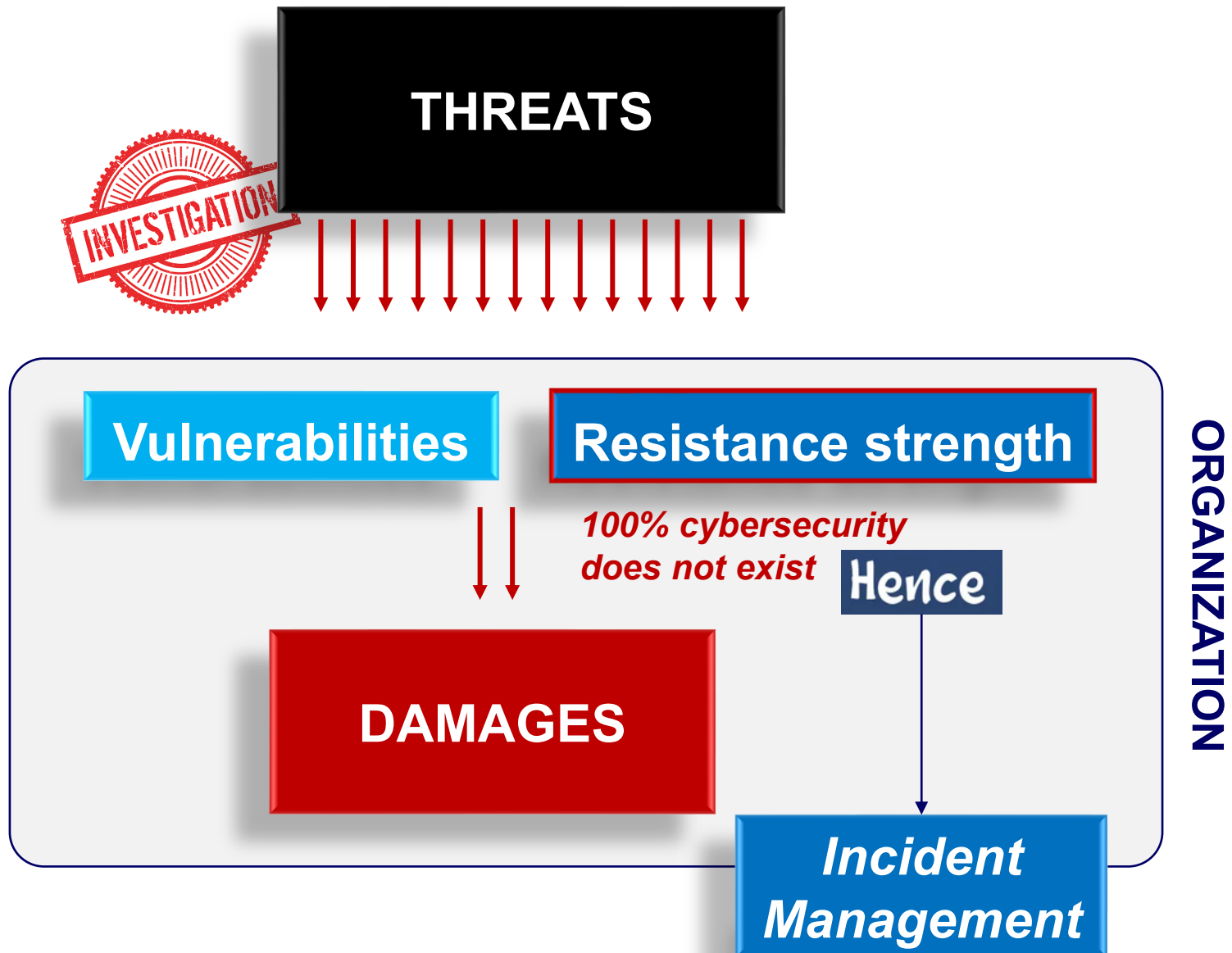


**DAMAGES**

**ORGANIZATION**



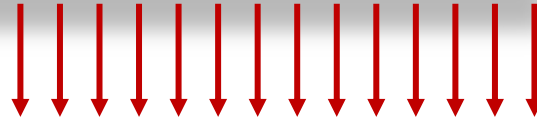
# The entire “cybersecurity game” is here



# This course is about *cyber defenses* (no *offensive*)

THREATS

We study **threat capability** (skill and resources) of the *cyber threat actor* (or *attacker*, never hacker)



VULNERABILITIES

Resistance strength



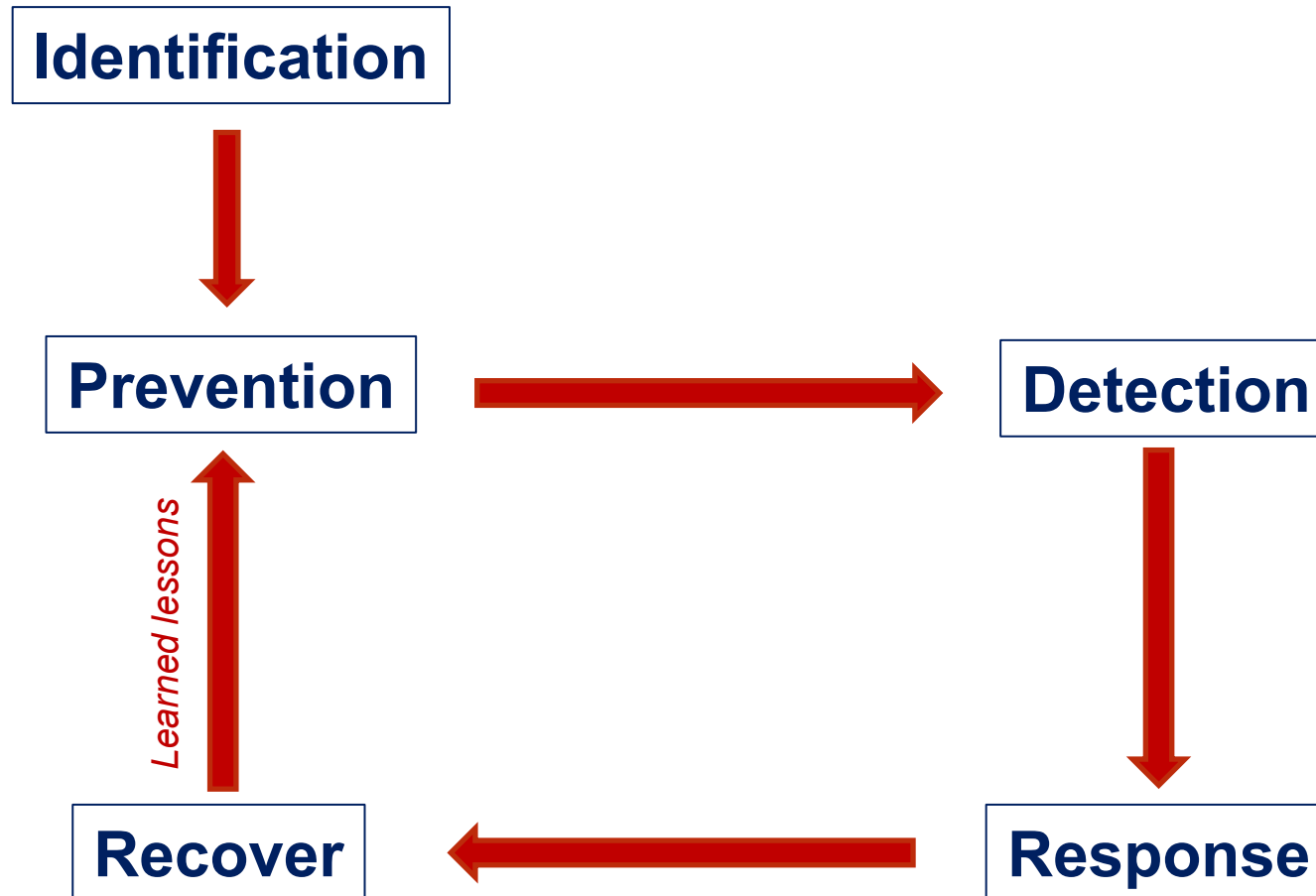
DAMAGES

ORGANIZATION

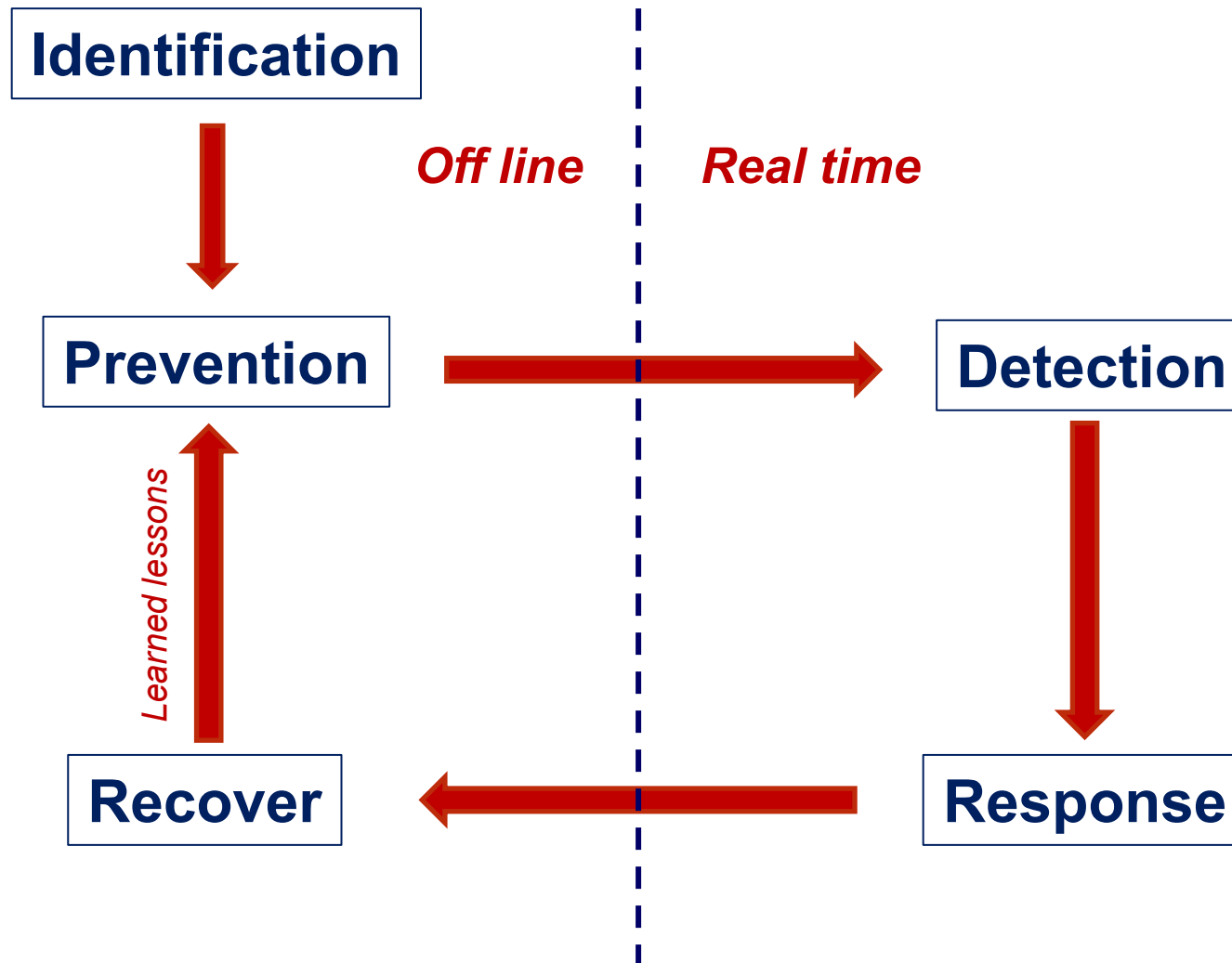
*Incident  
Management*



# The main phases of cyber defense



# Two approaches and attitudes for cyber defense



# NOT in Agenda – *What this course is not about*

1. Penetration testing
2. Exploiting software vulnerabilities
3. Malware analysis
4. Digital forensics

See other courses:

- “*Sicurezza dell’Informazione*”, Rebecca Montanari
- “*Amministazione di sistemi and Laboratorio di sicurezza*”, M. Prandini
- “*Digital forensics*”, A. Amoroso



# You will learn how to

- Identify the most important **resources** to be protected
- Identify the most likely **adversaries**
- Identify the most likely **damages**



Subject to the most important **constraints**:



- Money
- Time
- Law
- Competences (*lack of*)



**1<sup>st</sup> main focus  
of the course**

And to choose the **best cybersecurity defenses**  
in terms of methods, policies and technologies

**2<sup>nd</sup> main focus  
of the course**



# Cybersecurity spans across cultures

**Cybersecurity** is a  
**cognitive,**  
**technological,** and  
**management** problem

The real problem with cybersecurity is that we have *Paleolithic emotions, medieval institutions, and god-like technologies*

If you consider only one aspect, you have lost  
If you use a *one-fit-all* approach, you have lost  
→ There are not two equal companies, hence  
cybersecurity requires a tailor-made adaptive approach



Analytical  
gaze



# Typical vision on cybersecurity

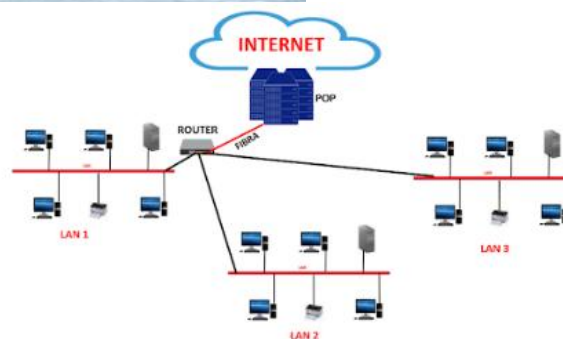


# The typical vision on cybersecurity is totally wrong!

## *Why?*



# Real corporate computer system

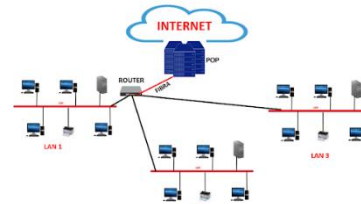
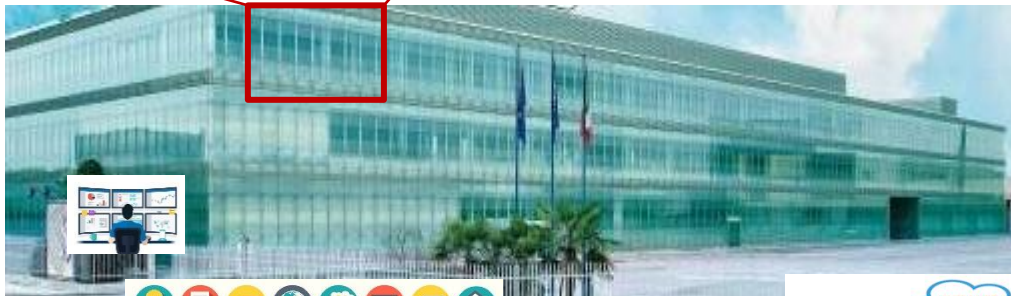


# Vulnerabilities of an organization



Email (>1)  
Linkedin, Facebook  
Messenger, WhatsApp  
Internet Service Provider  
Online banking, Amazon  
Meet, Zoom, Teams  
Dropbox, Azure 365, Google Docs  
AirNnB, Expedia, Hotels, Lastminute  
Trains, Airlines

**PC – Laptop  
Smartphone**



# Cybersecurity is characterized by a fascinating *dual nature*

- On one hand, cybersecurity is a sector in **constant flux** (with new vulnerabilities, new attack vectors, and new tools emerging every week)
- On the other hand, it rests on **fundamental principles** that remain unchanged and have stayed remarkably stable for decades

Main concepts are half a century old (dating back to Saltzer and Schroeder's 1975 paper, "The Protection of Information in Computer Systems"), yet they remain perfectly valid



# Fundamental principles

- 1) **Threat and risk analysis models** (*cyber risk management*): the way we reason about “who the attacker is, what they want, and what the attack surfaces are” remains the same, regardless of the technology.
- 2) **Fundamental vulnerability classes: *user-related vulnerabilities*, *software-related vulnerabilities*** (buffer overflows, code injection, race conditions, memory management issues) remain the same. The languages in which they manifest may change, but the underlying logic does not. The same for ***system* and *management vulnerabilities***.
- 3) **Cryptography, its mathematical primitives (RSA, elliptic curves, hashes), and associated protocols**: implementations change, but the underlying theory does not. [With the exception of *quantum-safe cryptography*.]
- 4) **Secure design principles: least privilege, strong authentication, defense-in-depth, and fail-safe defaults** (explicit permissions rather than default ones) remain perfectly valid in any architecture.



# Don't forget that Cybersecurity includes

- **Strong ethical dimension**
- **Legal aspects**
- **International standards and European norms**
- **Management dimension**
- **Practice**



# Calendar - *Exams*

**December 2026:** approximately **15, 16, 22, 23** (same hours and room of the class)

 **Approximately 25/30 out-of-town students take precedence over the choice of date (Send an email to me by November 7<sup>th</sup>) +**

*Approx. week* **January 11, 2027**

*Approx. week* **February 15, 2027**

**From March 2024 onwards: *just upon request***

**Important note: for the December pre-exams, you must register on *Alma esami official Web site***



# Type - *Exams*

**Oral on the fundamental principles that each of you must know  
+  
Possible discussion on solution(s) to a specific architecture**

- **Exam may be given in Italian or English**



# Last (or *first*) problem. **Let us** decide the rule in class



- **I will not use. You will not use**
- If any of you is aware of being *phone-alcoholic*, *social-alcoholic*, I can reduce length of each period of lesson to 30 minutes to give you the possibility to check your email, WA, social, ...
- Other suggestions are welcome

**Unacceptable:** students minding their businesses in class

**Even worse:** to distract colleagues

***It is not mandatory to attend my classes.  
If you come here, you must respect the rules***

